

HW3

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Q1: 1.3

(i)

Show that $\mathcal{F}_6 = \{A \subset \Omega : A \text{ is countable or } A^c \text{ is countable}\}$ from Example 1.1.4 is a σ -algebra.

Answer

Proof. Let Ω be a nonempty set.

- (a) $\Omega^c = \emptyset$, which is countable (even if Ω itself is not countable). Thus, $\Omega \in \mathcal{F}_6$.
- (b) Let $A \in \mathcal{F}_6$ be given as defined above.

Case 1: A is countable. Then $(A^c)^c = A$ is countable, so $A^c \in \mathcal{F}_6$.

Case 2: A^c is countable. Then $A^c \in \mathcal{F}_6$ by the definition of $\mathcal{F}_6$.

In either case, $A^c \in \mathcal{F}_6$.

- (d) Let $\{A_n\}_{n \geq 1} \subset \mathcal{F}_6$ be given, and set $A \equiv \bigcup_{n \geq 1} A_n$.

Case 1: A_n is countable for all $n \geq 1$. Then A is a countable union of countable sets, and hence is itself countable. Thus, $A \in \mathcal{F}_6$.

Case 2: A_{n_0} is uncountable for some $n_0 \in \mathbb{N}$. Since $A_{n_0} \in \mathcal{F}_6$, $A_{n_0}^c$ must be countable. Then, via De Morgan's law,

$$A^c = \left(\bigcup_{n \geq 1} A_n \right)^c = \bigcap_{n \geq 1} A_n^c \subset A_{n_0}^c.$$

Since any subset of a countable set is countable, A^c is countable, such that $A \in \mathcal{F}_6$.

- (c) By (a) and (b), $\emptyset = \Omega^c \in \mathcal{F}_6$. Let $A, B \in \mathcal{F}_6$ be given, and take $A_1 = A$, $A_2 = B$, and $A_n = \emptyset$ for all $n \geq 3$. Then by (d), $A \cup B = \bigcup_{n \geq 1} A_n \in \mathcal{F}_6$.

Thus, by (a)–(c), $\mathcal{F}_6$ is an algebra; and furthermore, by (d), $\mathcal{F}_6$ is a σ -algebra. //

(ii)

Define μ on $\mathcal{F}_6$ as $\mu(A) = 1$ if A is uncountable and $\mu(A) = 0$ if A is countable. Is μ a measure on $\mathcal{F}_6$? Justify.

Answer

Yes, μ is a measure on $\mathcal{F}_6$.

Proof.

- (a) $\mu(A) \in \{0, 1\} \subset [0, \infty]$ for all $A \in \mathcal{F}_6$, so property (a) of Definition 1.2.1 holds.

(b) $\emptyset$ is countable, so $\mu(\emptyset) = 0$, and property (b) of Definition 1.2.1 holds.

(c) *Claim:* If $A, B \in \mathcal{F}_6$ with $A \cap B = \emptyset$, then A and B are not both uncountable.

Suppose A is uncountable. Since $A \in \mathcal{F}_6$, A^c is countable. Since $A \cap B = \emptyset$, $B \subset A^c$, and so B is countable.

Now let $\{A_n\}_{n \geq 1} \subset \mathcal{F}_6$ be a disjoint collection, and let $A \equiv \bigcup_{n \geq 1} A_n$. Note that $A \in \mathcal{F}_6$ by (i). By the claim, *at most one* A_n is uncountable.

Case 1: A_n is countable for all $n \geq 1$. Then A is countable, and

$$\mu(A) = 0 = \sum_{n=1}^{\infty} 0 = \sum_{n=1}^{\infty} \mu(A_n).$$

Case 2: Exactly one A_{n_0} is uncountable. Then $A \supset A_{n_0}$ is uncountable, so $\mu(A) = 1$. Also, $\mu(A_n) = 0$ for all $n \neq n_0$, so

$$\sum_{n=1}^{\infty} \mu(A_n) = \mu(A_{n_0}) + \sum_{n \neq n_0} \mu(A_n) = 1 + 0 = 1 = \mu(A).$$

In either case, property (c) holds, and $(\Omega, \mathcal{F}_6, \mu)$ is a measure space. //

Q2: 1.8

Let $\Omega \equiv \{1, 2, \dots\} = \mathbb{N}$ and $A_i \equiv \{j : j \in \mathbb{N}, j \geq i\}$, $i \in \mathbb{N}$. Show that $\sigma\langle \mathcal{A} \rangle = \mathcal{P}(\Omega)$ where $\mathcal{A} = \{A_i : i \in \mathbb{N}\}$.

Answer

Proof.

($\subset$): $\sigma\langle \mathcal{A} \rangle \subset \mathcal{P}(\Omega)$.

$A_i \subset \Omega$ for all $i \in \mathbb{N}$, so $\mathcal{A} \subset \mathcal{P}(\Omega)$. Also, $\mathcal{P}(\Omega)$ is a σ -algebra (Example 1.1.2). Thus $\mathcal{P}(\Omega) \in \mathcal{I}(\mathcal{A})$, and so

$$\sigma\langle \mathcal{A} \rangle = \bigcap_{\mathcal{F} \in \mathcal{I}(\mathcal{A})} \mathcal{F} \subset \mathcal{P}(\Omega).$$

($\supset$): $\sigma\langle \mathcal{A} \rangle \supset \mathcal{P}(\Omega)$.

Fix $i \in \mathbb{N}$. For $j \in \mathbb{N}$,

$$j \in A_i \cap A_{i+1}^c \iff i \leq j < i+1 \iff j = i,$$

so $\{i\} = A_i \cap A_{i+1}^c$. Since $A_i, A_{i+1} \in \mathcal{A} \subset \sigma\langle \mathcal{A} \rangle$ and $\sigma\langle \mathcal{A} \rangle$ is closed under complements and countable unions,

$$\{i\} = (A_i^c \cup A_{i+1})^c \in \sigma\langle \mathcal{A} \rangle.$$

Now let $B \in \mathcal{P}(\Omega)$. Since $B \subset \mathbb{N}$, B is countable, so

$$B = \bigcup_{i \in B} \{i\}$$

is a countable union of sets in $\sigma\langle \mathcal{A} \rangle$. Hence $B \in \sigma\langle \mathcal{A} \rangle$. (If $B = \emptyset$, the union is empty, and trivially $\emptyset = \Omega^c \in \sigma\langle \mathcal{A} \rangle$.)

Thus, $\sigma\langle \mathcal{A} \rangle = \mathcal{P}(\Omega)$. //

Q3: 1.10

Show that in Example 1.1.6, $\mathcal{O}_j \subset \sigma\langle\mathcal{O}_i\rangle$ for all $1 \leq i, j \leq 4$.

(Okay to do for $k = 2$, i.e. $\mathbb{R}^2$.)

Note on General Strategy: If $\mathcal{O}_A \subset \sigma\langle\mathcal{O}_B\rangle$, then $\sigma\langle\mathcal{O}_A\rangle \subset \sigma\langle\mathcal{O}_B\rangle$, since $\sigma\langle\mathcal{O}_B\rangle$ is a σ -algebra containing $\mathcal{O}_A$. So it suffices to establish the cycle

$$\mathcal{O}_1 \subset \sigma\langle\mathcal{O}_3\rangle, \quad \mathcal{O}_3 \subset \sigma\langle\mathcal{O}_4\rangle, \quad \mathcal{O}_4 \subset \sigma\langle\mathcal{O}_2\rangle, \quad \mathcal{O}_2 \subset \sigma\langle\mathcal{O}_1\rangle.$$

(i)

$$\mathcal{O}_1 = \{(a_1, b_1) \times (a_2, b_2) : -\infty \leq a_1 < b_1 \leq \infty, -\infty \leq a_2 < b_2 \leq \infty\}$$

Answer

Proof. Let $R = (a_1, b_1) \times (a_2, b_2) \in \mathcal{O}_1$, and let

$$J \equiv \{(p_1, q_1, p_2, q_2) \in \mathbb{Q}^4 : a_i < p_i < q_i < b_i, i = 1, 2\}.$$

We claim

$$R = \bigcup_{(p_1, q_1, p_2, q_2) \in J} (p_1, q_1) \times (p_2, q_2).$$

($\supset$) Since $a_i < p_i < q_i < b_i$, each rectangle in the union is contained in R .

($\subset$) Let $(x_1, x_2) \in R$. By the density of $\mathbb{Q}$, choose $p_i \in (a_i, x_i) \cap \mathbb{Q}$ and $q_i \in (x_i, b_i) \cap \mathbb{Q}$ for $i = 1, 2$. Then $(p_1, q_1, p_2, q_2) \in J$ and $(x_1, x_2) \in (p_1, q_1) \times (p_2, q_2)$.

Each $(p_1, q_1) \times (p_2, q_2)$ has rational endpoints, so it lies in $\mathcal{O}_3$, and $J \subset \mathbb{Q}^4$ is countable. Hence R is a countable union of sets in $\sigma\langle\mathcal{O}_3\rangle$, so $R \in \sigma\langle\mathcal{O}_3\rangle$, and we conclude $\mathcal{O}_1 \subset \sigma\langle\mathcal{O}_3\rangle$. //

(ii)

$$\mathcal{O}_2 = \{(-\infty, x_1) \times (-\infty, x_2) : -\infty < x_1, x_2 < \infty\}$$

Answer

Proof. Let $(-\infty, x_1) \times (-\infty, x_2) \in \mathcal{O}_2$. Taking $a_i = -\infty$ and $b_i = x_i$ in the definition of $\mathcal{O}_1$ (valid since $-\infty < x_i < \infty$) shows this set is in $\mathcal{O}_1$. Thus $\mathcal{O}_2 \subset \mathcal{O}_1 \subset \sigma\langle\mathcal{O}_1\rangle$. //

(iii)

$$\mathcal{O}_3 = \{(a_1, b_1) \times (a_2, b_2) : a_1 < b_1, a_2 < b_2, a_1, a_2, b_1, b_2 \in \mathbb{Q}\}$$

Answer

Proof. Let $R = (a_1, b_1) \times (a_2, b_2) \in \mathcal{O}_3$, and let $c_{i,n} \equiv a_i + n^{-1} \in \mathbb{Q}$ for $n \in \mathbb{N}$. We claim

$$R = \bigcup_{n \geq 1} [c_{1,n}, b_1) \times [c_{2,n}, b_2).$$

($\supset$) Since $c_{i,n} > a_i$, each set in the union is contained in R .

($\subset$) Let $y \in R$, and choose n with $n^{-1} < \min_i(y_i - a_i)$. Then $c_{i,n} < y_i < b_i$ for $i = 1, 2$, so $y \in [c_{1,n}, b_1) \times [c_{2,n}, b_2)$.

Since $\sigma\langle\mathcal{O}_4\rangle$ is closed under countable unions, it suffices to show $[c_1, b_1) \times [c_2, b_2) \in \sigma\langle\mathcal{O}_4\rangle$ for $c_1, c_2 \in \mathbb{Q}$. Let $H(x_1, x_2) \equiv (-\infty, x_1) \times (-\infty, x_2)$, so that $\mathcal{O}_4 = \{H(x_1, x_2) : x_1, x_2 \in \mathbb{Q}\}$.

Since a point $y \in H(b_1, b_2)$ lies outside $H(c_1, b_2)$ exactly when $y_1 \geq c_1$, and outside $H(b_1, c_2)$ exactly when $y_2 \geq c_2$, we have

$$[c_1, b_1) \times [c_2, b_2) = H(b_1, b_2) \setminus [H(c_1, b_2) \cup H(b_1, c_2)].$$

Since $b_i, c_i \in \mathbb{Q}$, each of $H(b_1, b_2)$, $H(c_1, b_2)$, and $H(b_1, c_2)$ is in $\mathcal{O}_4$. Because $\sigma\langle\mathcal{O}_4\rangle$ is closed under finite unions and differences, $[c_1, b_1) \times [c_2, b_2) \in \sigma\langle\mathcal{O}_4\rangle$.

Hence $R \in \sigma\langle\mathcal{O}_4\rangle$, and so $\mathcal{O}_3 \subset \sigma\langle\mathcal{O}_4\rangle$. //

(iv)

$$\mathcal{O}_4 = \{(-\infty, x_1) \times (-\infty, x_2) : x_1, x_2 \in \mathbb{Q}\}$$

Answer

Proof. We show $\mathcal{O}_4 \subset \sigma\langle\mathcal{O}_2\rangle$. Since $\mathbb{Q} \subset \mathbb{R}$, $\mathcal{O}_4 \subset \mathcal{O}_2 \subset \sigma\langle\mathcal{O}_2\rangle$. //

Conclusion

By (i), (iii), (iv), and (ii), in that order, and the note on general strategy of proof,

$$\sigma\langle\mathcal{O}_1\rangle \subset \sigma\langle\mathcal{O}_3\rangle \subset \sigma\langle\mathcal{O}_4\rangle \subset \sigma\langle\mathcal{O}_2\rangle \subset \sigma\langle\mathcal{O}_1\rangle.$$

So all four σ -algebras are equal, and $\mathcal{O}_j \subset \sigma\langle\mathcal{O}_j\rangle = \sigma\langle\mathcal{O}_i\rangle$ for all $1 \leq i, j \leq 4$. //

Q4: 1.18

Let $\Omega \equiv \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and $A_n = \{j : j \in \mathbb{N}, j \geq n\}$, $n \in \mathbb{N}$. Let μ be the counting measure on $(\Omega, \mathcal{F})$. Verify that $\lim_{n \rightarrow \infty} \mu(A_n) \neq \mu(\bigcap_{n \geq 1} A_n)$.

Answer

Proof. For each $n \in \mathbb{N}$, the map $j \mapsto n+j-1$ is an injection from $\mathbb{N}$ into A_n , so A_n is infinite and $\mu(A_n) = \infty$. Hence $\lim_{n \rightarrow \infty} \mu(A_n) = \infty$.

On the other hand, $\bigcap_{n \geq 1} A_n = \emptyset$. For any $j \in \mathbb{N}$, $j < j+1$ gives $j \notin A_{j+1}$, so $j \notin \bigcap_{n \geq 1} A_n$. By property (b) of Definition 1.2.1, $\mu(\bigcap_{n \geq 1} A_n) = \mu(\emptyset) = 0$. Therefore

$$\lim_{n \rightarrow \infty} \mu(A_n) = \infty \neq 0 = \mu\left(\bigcap_{n \geq 1} A_n\right). //$$

Q5: 1.19

Let Ω be a nonempty set and let $\mathcal{C} \subset \mathcal{P}(\Omega)$ be a semialgebra. Let

$$\mathcal{A}(\mathcal{C}) \equiv \left\{ A : A = \bigcup_{i=1}^k B_i : B_i \in \mathcal{C}, i = 1, 2, \dots, k, k \in \mathbb{N} \right\}$$

(i)

Show that $\mathcal{A}(\mathcal{C})$ is the smallest algebra containing $\mathcal{C}$.

Answer

Proof. We assume $\mathcal{C} \neq \emptyset$; otherwise $\mathcal{A}(\mathcal{C}) = \emptyset$, which is not an algebra.

Each $B \in \mathcal{C}$ is in $\mathcal{A}(\mathcal{C})$ (take $k = 1$, $B_1 = B$), so $\mathcal{C} \subset \mathcal{A}(\mathcal{C})$. We next check that $\mathcal{A}(\mathcal{C})$ is an algebra.

- (a) Pick any $A \in \mathcal{C}$. Since $\mathcal{C}$ is a semialgebra, $A^c = \bigcup_{i=1}^m B_i$ for some $B_1, \dots, B_m \in \mathcal{C}$. Hence $\Omega = A \cup B_1 \cup \dots \cup B_m \in \mathcal{A}(\mathcal{C})$.
- (b) Let $A = \bigcup_{i=1}^k B_i$ with $B_i \in \mathcal{C}$, and write each $B_i^c = \bigcup_{j=1}^{n_i} B_{ij}$ with $B_{ij} \in \mathcal{C}$. Then

$$A^c = \bigcap_{i=1}^k B_i^c = \bigcap_{i=1}^k \bigcup_{j=1}^{n_i} B_{ij} = \bigcup_{j_1=1}^{n_1} \dots \bigcup_{j_k=1}^{n_k} \left(\bigcap_{i=1}^k B_{ij_i} \right),$$

where the last equality holds because a point lies in $\bigcap_{i=1}^k \bigcup_{j=1}^{n_i} B_{ij}$ exactly when, for each i , it lies in some B_{ij_i} . Since $\mathcal{C}$ is closed under pairwise intersections, and hence by induction under finite intersections, each $\bigcap_{i=1}^k B_{ij_i} \in \mathcal{C}$. Thus A^c is a union of $n_1 n_2 \dots n_k$ sets in $\mathcal{C}$, so $A^c \in \mathcal{A}(\mathcal{C})$.

- (c) Let $A = \bigcup_{i=1}^k B_i$ and $D = \bigcup_{j=1}^m D_j$ be in $\mathcal{A}(\mathcal{C})$. Then $A \cup D = B_1 \cup \dots \cup B_k \cup D_1 \cup \dots \cup D_m$ is a union of $k + m$ sets in $\mathcal{C}$, so $A \cup D \in \mathcal{A}(\mathcal{C})$.

By (a)–(c), $\mathcal{A}(\mathcal{C})$ is an algebra.

Finally, let $\mathcal{E}$ be any algebra with $\mathcal{C} \subset \mathcal{E}$, and let $A = \bigcup_{i=1}^k B_i \in \mathcal{A}(\mathcal{C})$. Each $B_i \in \mathcal{E}$, and $\mathcal{E}$ is closed under finite unions (by induction on pairwise unions), so $A \in \mathcal{E}$. Thus $\mathcal{A}(\mathcal{C}) \subset \mathcal{E}$, and $\mathcal{A}(\mathcal{C})$ is the smallest algebra containing $\mathcal{C}$. //

(ii)

Show also that $\sigma\langle \mathcal{C} \rangle = \sigma\langle \mathcal{A}(\mathcal{C}) \rangle$.

Answer

Proof. ($\subset$) By (i), $\mathcal{C} \subset \mathcal{A}(\mathcal{C}) \subset \sigma\langle \mathcal{A}(\mathcal{C}) \rangle$. Since $\sigma\langle \mathcal{A}(\mathcal{C}) \rangle$ is a σ -algebra containing $\mathcal{C}$, $\sigma\langle \mathcal{C} \rangle \subset \sigma\langle \mathcal{A}(\mathcal{C}) \rangle$.

($\supset$) $\sigma\langle \mathcal{C} \rangle$ is a σ -algebra, hence an algebra, containing $\mathcal{C}$, so $\mathcal{A}(\mathcal{C}) \subset \sigma\langle \mathcal{C} \rangle$ by (i). Since $\sigma\langle \mathcal{C} \rangle$ is a σ -algebra containing $\mathcal{A}(\mathcal{C})$, $\sigma\langle \mathcal{A}(\mathcal{C}) \rangle \subset \sigma\langle \mathcal{C} \rangle$.

Combining both relations, we conclude $\sigma\langle \mathcal{C} \rangle = \sigma\langle \mathcal{A}(\mathcal{C}) \rangle$. //

Q6: 1.20

Let μ^* be as in (3.1) of Section 1.3:

Given a measure μ on a semialgebra $\mathcal{C} \subset \mathcal{P}(\Omega)$, the outer measure induced by μ is the function μ^* , defined on $\mathcal{P}(\Omega)$ as:

$$\mu^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n \geq 1} \subset \mathcal{C}, A \subset \bigcup_{n \geq 1} A_n \right\}$$

(Hint: Fix $0 < \epsilon < \infty$. If $\mu^*(A_n) < \infty$ for all $n \in \mathbb{N}$, then find, for each n , a cover $\{A_{nj}\}_{j \geq 1} \subset \mathcal{C}$, $A_n \subset \bigcup_{j \geq 1} A_{nj}$ such that $\sum_{j=1}^{\infty} \mu(A_{nj}) \leq \mu^*(A_n) + \frac{\epsilon}{2^n}$.)

Verify:

(3.4)

$$\mu^*(\emptyset) = 0$$

Answer

Notation. For $A \subset \Omega$, let

$$\Sigma(A) \equiv \left\{ \sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n \geq 1} \subset \mathcal{C}, A \subset \bigcup_{n \geq 1} A_n \right\},$$

so that $\mu^*(A) = \inf \Sigma(A)$. Since μ takes values in $[0, \infty]$, every element of $\Sigma(A)$ is nonnegative, so $\mu^*(A) \geq 0$ for all $A \subset \Omega$.

Proof. We first show $\emptyset \in \mathcal{C}$. Assume $\mathcal{C} \neq \emptyset$, as in Q5, and pick any $A \in \mathcal{C}$. Since $\mathcal{C}$ is a semialgebra, $A^c = \bigcup_{i=1}^k B_i$ for some $B_1, \dots, B_k \in \mathcal{C}$, and $A \cap B_1 \in \mathcal{C}$ since $\mathcal{C}$ is closed under intersections. But $A \cap B_1 \subset A \cap A^c = \emptyset$, so $\emptyset = A \cap B_1 \in \mathcal{C}$.

Now take $A_n = \emptyset$ for all $n \geq 1$. This sequence lies in $\mathcal{C}$ and covers $\emptyset$, and $\mu(\emptyset) = 0$ since μ is a measure on $\mathcal{C}$. Thus

$$0 \leq \mu^*(\emptyset) \leq \sum_{n=1}^{\infty} \mu(\emptyset) = 0,$$

and so $\mu^*(\emptyset) = 0$. //

(3.5)

$$A \subset B \Rightarrow \mu^*(A) \leq \mu^*(B)$$

Answer

Proof. Suppose $A \subset B$, and let $\{A_n\}_{n \geq 1} \subset \mathcal{C}$ with $B \subset \bigcup_{n \geq 1} A_n$. Then $A \subset B \subset \bigcup_{n \geq 1} A_n$, so $\sum_{n=1}^{\infty} \mu(A_n) \in \Sigma(A)$. Thus $\Sigma(B) \subset \Sigma(A)$.

The infimum over a larger set is no greater than the infimum over a smaller one, so

$$\mu^*(A) = \inf \Sigma(A) \leq \inf \Sigma(B) = \mu^*(B). \quad //$$

(3.6)

For any $\{A_n\}_{n \geq 1} \subset \mathcal{P}(\Omega)$,

$$\mu^*\left(\bigcup_{n \geq 1} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n).$$

Answer

Proof. Let $\{A_n\}_{n \geq 1} \subset \mathcal{P}(\Omega)$ be given, and set $A \equiv \bigcup_{n \geq 1} A_n$. If $\mu^*(A_n) = \infty$ for some n , the right-hand side is ∞ and there is nothing to prove. So suppose $\mu^*(A_n) < \infty$ for all n , and fix $0 < \epsilon < \infty$.

For each n , $\mu^*(A_n) + \epsilon/2^n$ exceeds $\inf \Sigma(A_n)$, so it is not a lower bound of $\Sigma(A_n)$. Hence there is a cover $\{A_{nj}\}_{j \geq 1} \subset \mathcal{C}$ of A_n with

$$\sum_{j=1}^{\infty} \mu(A_{nj}) \leq \mu^*(A_n) + \frac{\epsilon}{2^n}.$$

The collection $\{A_{nj} : n, j \geq 1\} \subset \mathcal{C}$ is indexed by $\mathbb{N} \times \mathbb{N}$, so it can be enumerated as a single sequence in $\mathcal{C}$. It covers A , since $A = \bigcup_{n \geq 1} A_n \subset \bigcup_{n \geq 1} \bigcup_{j \geq 1} A_{nj}$. Because the terms $\mu(A_{nj})$ are nonnegative, their sum does not depend on the order of summation, so

$$\mu^*(A) \leq \sum_{n=1}^{\infty} \sum_{j=1}^{\infty} \mu(A_{nj}) \leq \sum_{n=1}^{\infty} \left(\mu^*(A_n) + \frac{\epsilon}{2^n} \right) = \sum_{n=1}^{\infty} \mu^*(A_n) + \epsilon.$$

Since $\epsilon > 0$ is arbitrary, $\mu^*(A) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$. //

Q7: 1.24

Verify that $\mathcal{C}_2$, defined in (3.11), is a semialgebra.

Answer

Recall from (3.7) and (3.11):

$$\mathcal{C} \equiv \{(a, b] : -\infty \leq a \leq b < \infty\} \cup \{(a, \infty) : -\infty \leq a < \infty\}, \quad \mathcal{C}_2 \equiv \{I_1 \times I_2 : I_1, I_2 \in \mathcal{C}\}.$$

Proof. A class is a semialgebra if it is closed under pairwise intersections and the complement of each member is a finite disjoint union of members. We verify both properties for $\mathcal{C}$, then use them to verify both for $\mathcal{C}_2$.

Let $-\infty \leq a, c$ and $b, d < \infty$. Reading $(u, v]$ as $\emptyset$ when $u > v$, which is harmless since $\emptyset = (0, 0] \in \mathcal{C}$, we have

$$(a, b] \cap (c, d] = (a \vee c, b \wedge d], \quad (a, b] \cap (c, \infty) = (a \vee c, b], \quad (a, \infty) \cap (c, \infty) = (a \vee c, \infty),$$

so $\mathcal{C}$ is closed under intersections. For complements,

$$(a, b]^c = (-\infty, a] \cup (b, \infty), \quad (a, \infty)^c = (-\infty, a],$$

each a finite disjoint union of members of $\mathcal{C}$. Here $(-\infty, a] \in \mathcal{C}$ because (3.7) permits a left endpoint of $-\infty$; when $a = -\infty$, it is $\emptyset$.

Now let $A = I_1 \times I_2$ and $B = I_3 \times I_4$ be in $\mathcal{C}_2$. A point (x, y) lies in both exactly when $x \in I_1 \cap I_3$ and $y \in I_2 \cap I_4$, so

$$A \cap B = (I_1 \cap I_3) \times (I_2 \cap I_4).$$

Since $\mathcal{C}$ is closed under intersections, $I_1 \cap I_3$ and $I_2 \cap I_4$ are in $\mathcal{C}$, and hence $A \cap B \in \mathcal{C}_2$.

For complements, let $A = I_1 \times I_2 \in \mathcal{C}_2$. Since $\mathcal{C}$ is a semialgebra, we can write $I_1^c = \bigcup_{j=1}^{n_1} I_{1j}$ and $I_2^c = \bigcup_{l=1}^{n_2} I_{2l}$ as disjoint unions of members of $\mathcal{C}$. Setting $I_{10} \equiv I_1$ and $I_{20} \equiv I_2$, the sets $I_{10}, \dots, I_{1n_1}$ are pairwise disjoint with union $\mathbb{R}$, as are $I_{20}, \dots, I_{2n_2}$. It follows that the products $I_{1j} \times I_{2l}$, for $0 \leq j \leq n_1$ and $0 \leq l \leq n_2$, are pairwise disjoint with union $\mathbb{R}^2$: each (x, y) lies in exactly one of them, determined by which I_{1j} contains x and which I_{2l} contains y . Removing $A = I_{10} \times I_{20}$ leaves

$$A^c = \bigcup_{(j,l) \neq (0,0)} I_{1j} \times I_{2l},$$

a disjoint union of $(n_1 + 1)(n_2 + 1) - 1$ members of $\mathcal{C}_2$. Thus $\mathcal{C}_2$ is a semialgebra. //

Q8: Extra

Consider the following function:

$$F(x) = x \mathbb{I}\{0 < x < 1\} + \mathbb{I}\{x \geq 1\}, \quad x \in \mathbb{R}$$

(i)

Is this $F(\cdot)$ non-decreasing, right-continuous? Check.

Answer

Yes. In fact, F is non-decreasing and continuous, and hence right-continuous.

Proof. We claim $F(x) = \min\{\max\{x, 0\}, 1\}$ for all $x \in \mathbb{R}$. If $x \leq 0$, then $\max\{x, 0\} = 0$, so the right side is $0 = F(x)$. If $0 < x < 1$, then $\max\{x, 0\} = x < 1$, so the right side is $x = F(x)$. If $x \geq 1$, then $\max\{x, 0\} = x \geq 1$, so the right side is $1 = F(x)$.

Let $g(x) \equiv \max\{x, 0\} = \frac{x+|x|}{2}$ and $h(y) \equiv \min\{y, 1\} = \frac{y+1-|y-1|}{2}$. Both are continuous, since $|\cdot|$ is continuous. Both are also non-decreasing, since for $x_1 \leq x_2$ we have $\max\{x_1, 0\} \leq \max\{x_2, 0\}$ and $\min\{x_1, 1\} \leq \min\{x_2, 1\}$. Thus $F = h \circ g$ is non-decreasing and continuous, as a composition of non-decreasing, continuous functions. In particular, $F(x+) = F(x)$ for all $x \in \mathbb{R}$, so F is right-continuous.
//

(ii)

Define suitably a Lebesgue-Stieltjes measure μ_F on a suitable semialgebra $\mathcal{C}$. Do not have to prove $\mathcal{C}$ is a semi-algebra or μ_F is a measure.

Answer

Take $\mathcal{C}$ to be the semialgebra in (3.7):

$$\mathcal{C} \equiv \{(a, b] : -\infty \leq a \leq b < \infty\} \cup \{(a, \infty) : -\infty \leq a < \infty\}.$$

Set $F(-\infty) \equiv \lim_{x \rightarrow -\infty} F(x) = 0$ and $F(\infty) \equiv \lim_{x \rightarrow \infty} F(x) = 1$. Following (3.8), define

$$\mu_F((a, b]) = F(b+) - F(a+) = F(b) - F(a), \quad \mu_F((a, \infty)) = F(\infty) - F(a+) = 1 - F(a),$$

where $F(x+) = F(x)$ by (i).

(iii)

Prove that $\sigma\langle \mathcal{C} \rangle = \mathcal{B}(\mathbb{R})$. (Can use the fact that $\mathcal{B}(\mathbb{R}) = \sigma\langle \{\text{open sets}\} \rangle = \sigma\langle \mathcal{O}_1 \rangle = \sigma\langle \mathcal{O}_2 \rangle = \sigma\langle \mathcal{O}_3 \rangle = \sigma\langle \mathcal{O}_4 \rangle$ as in 1.10.)

Answer

Here, for $k = 1$, $\mathcal{O}_1 = \{(a, b) : -\infty \leq a < b \leq \infty\}$ and $\mathcal{O}_2 = \{(-\infty, x) : x \in \mathbb{R}\}$.

Proof. ($\subset$) It suffices to show $\mathcal{C} \subset \sigma\langle \mathcal{O}_1 \rangle = \mathcal{B}(\mathbb{R})$. Sets of the form (a, ∞) are in $\mathcal{O}_1$ directly (take $b = \infty$), and $(a, a] = \emptyset \in \mathcal{B}(\mathbb{R})$. For $-\infty \leq a < b < \infty$,

$$(a, b] = \bigcap_{n \geq 1} (a, b + n^{-1}),$$

since $a < y < b + n^{-1}$ for all n iff $a < y \leq b$. Each $(a, b + n^{-1}) \in \mathcal{O}_1$, and $\sigma\langle \mathcal{O}_1 \rangle$ is closed under countable intersections, so $(a, b] \in \sigma\langle \mathcal{O}_1 \rangle$. Thus $\mathcal{C} \subset \mathcal{B}(\mathbb{R})$, and hence $\sigma\langle \mathcal{C} \rangle \subset \mathcal{B}(\mathbb{R})$.

($\supset$) It suffices to show $\mathcal{O}_2 \subset \sigma\langle \mathcal{C} \rangle$. For $x \in \mathbb{R}$,

$$(-\infty, x) = \bigcup_{n \geq 1} (-\infty, x - n^{-1}],$$

since each set in the union lies in $(-\infty, x)$, and any $y < x$ satisfies $y \leq x - n^{-1}$ once $n^{-1} \leq x - y$. Each $(-\infty, x - n^{-1}] \in \mathcal{C}$ (take $a = -\infty$), so $(-\infty, x) \in \sigma\langle \mathcal{C} \rangle$. Thus $\mathcal{O}_2 \subset \sigma\langle \mathcal{C} \rangle$, and hence $\mathcal{B}(\mathbb{R}) = \sigma\langle \mathcal{O}_2 \rangle \subset \sigma\langle \mathcal{C} \rangle$.
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(iv)

Argue that there exist a μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ that extends μ_F defined above. Do not need to prove this step - just state a result for your answer.

Answer

Since μ_F is a measure on the semialgebra $\mathcal{C}$, Carathéodory's extension theorem (Theorem 1.3.3) gives that the outer measure μ_F^* induced by μ_F via (3.1) is a measure on the σ -algebra $\mathcal{M}_{\mu_F^*}$ (Theorem 1.3.2), with $\mathcal{C} \subset \mathcal{M}_{\mu_F^*}$ and $\mu_F^* = \mu_F$ on $\mathcal{C}$.

Since $\mathcal{M}_{\mu_F^*}$ is a σ -algebra containing $\mathcal{C}$, (iii) gives $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}) \subset \mathcal{M}_{\mu_F^*}$. Thus $\mu \equiv \mu_F^*|_{\mathcal{B}(\mathbb{R})}$ is a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with $\mu = \mu_F$ on $\mathcal{C}$.

(v)

Is this extension unique? (i.e. if μ' is another extension of μ_F , then is it true that $\mu(A) = \mu'(A)$, for all $A \in \mathcal{B}(\mathbb{R})$?). Justify your answer (no proof needed). What is the “common name” for this measure μ ?

Answer

Yes, the extension is unique.

Since $\mathbb{R} = (-\infty, \infty) \in \mathcal{C}$ and $\mu_F(\mathbb{R}) = F(\infty) - F(-\infty) = 1$, any extension μ' of μ_F satisfies $\mu'(\mathbb{R}) = \mu(\mathbb{R}) = 1$, so μ and μ' are both finite. Moreover, $\mathcal{C}$ is a π -system, since a semialgebra is closed under intersections, and $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$ by (iii). Since μ and μ' agree on $\mathcal{C}$, the uniqueness of measures (Theorem 1.2.4) gives $\mu = \mu'$ on $\mathcal{B}(\mathbb{R})$.

The measure μ is the Uniform(0, 1) distribution, i.e., the probability measure with CDF F . Equivalently, it is Lebesgue measure restricted to $(0, 1)$, i.e., $\mu(A) = m(A \cap (0, 1))$ for $A \in \mathcal{B}(\mathbb{R})$.